Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - IV

Course Code: BCA431 **Max Marks: 100**

Course Name: GRAPH THEORY **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Enhance research, inquiry and analytical thinking abilities.

CO2: To improve proof-writing skills.

CO3: Apply the basics of Graph Theory to solving practical problems.

Section A **3*10=30**

1)	Construct a schedule for three final exams using a graph, ensuring no two exams scheduled at the same time are taken by the same student and explain your choice of vertices and edges.	**3**	(CO1, L3)
2)	Construct a spanning tree for the given graph G, which consists of vertices {A, B, C, D, E} and edges {(A,B), (B,C), (C,D), (D,E), (E,A)}. Explain your steps.	**3**	(CO1, L3)
3) a)	Construct a graph that represents the following scenario: Three students are enrolled in two courses, each course is offered by one of three lecturers, and each lecturer teaches at least one course.	**3**	(CO2, L3)
	OR		
b)	Determine the number of edges in a simple graph with 8 vertices if every vertex has degree 3.	**3**	(CO2, L3)
4)	Examine how graph theory can be applied to optimize network reliability for a university campus network with specific focus on the placement of routers.	**3**	(CO3, L4)
5)	Explain the concept of a simple graph and describe how to calculate the number of edges in such a graph given its number of vertices.	**3**	(CO1, L2)
6)	Discuss how graph theory can be applied to optimize network routing in computer networks, focusing on the concept of shortest path algorithms. Provide an example of Dijkstra's algorithm to find the shortest path in a given weighted graph.	**3**	(CO1, L2)
7)	Discuss how the number of edges in a simple graph with n vertices can be determined if every vertex is connected to exactly three other vertices. Provide your answer briefly.	**3**	(CO2, L2)
8) a)	Determine whether the given graph is a tree and justify your answer by either providing a spanning tree or identifying a cycle.	**3**	(CO2, L3)
	OR		

b)	Construct a 4-ary tree with 20 vertices and determine the number of internal vertices in this tree.	**3**	(CO2, L3)
9)	Analyze the difference between an Eulerian and a Hamiltonian path/circuit. Provide an example of each.	**3**	(CO3, L4)
10)	Examine a given graph to determine if it contains an Eulerian circuit or a Hamiltonian cycle. If neither exists, provide reasons why not. [Graph: A simple undirected graph with 6 vertices and 8 edges, where each vertex has a degree of at least two.]	**3**	(CO1, L4)

Section B **7*3=21**

11)	Examine how graph theory can be applied to optimize network flow in a university campus where various departments require internet access with varying bandwidth requirements. Describe and analyze an algorithm that could be used to determine the maximum possible flow of data from the main server to all departments, considering the constraints on link capacities.	**7**	(CO3, L4)
12) a)	Analyze how the shortest path problem can be solved using Dijkstra's algorithm for the given graph G with 10 vertices and 15 edges. Calculate the total weight of the shortest path from vertex A to vertex J, and discuss any potential limitations or assumptions in your approach.	**7**	(CO3, L4)
	OR		
b)	Analyze how the structure of a 4-ary tree can be used to model and solve a practical problem involving resource allocation in a network of departments within an organization. Specifically, (i) construct a 4-ary tree with 15 vertices to represent this scenario, ensuring that each non-pendant vertex has exactly four children; (ii) determine the number of leaves and internal vertices in your constructed tree; (iii) explain how this tree structure can optimize resource allocation among departments.	**7**	(CO3, L4)
13)	Construct a general -ary tree with 15 vertices such that it has exactly 3 vertices of degree and 4 vertices of degree 1 (pendant vertices). Determine the total number of edges in this tree and verify if your construction satisfies the properties of an -ary tree. Additionally, compute how many internal vertices are present in your constructed tree.	**7**	(CO1, L3)

Section C **7*4=28**

14)	Discuss how graph theory can be applied to optimize university course scheduling. Specifically, address the following points: (i) Describe a method for creating a course schedule using graph coloring. (ii) Explain the process of identifying potential conflicts in the schedule and resolving them through vertex and edge manipulations. (iii) Provide an example of a simple graph representing a set of courses and their prerequisites, and demonstrate how to apply the scheduling algorithm.	**7**	(CO3, L2)
15)	A telecommunications company needs to optimize its data transmission strategy for five different symbols with given frequencies: (15%), (20%), (25%), (30%), and (10%). Construct a Huffman tree to encode these symbols. Calculate the average number of bits used to encode each symbol. Also, determine how many total bits would be saved compared to using equal-length codes for all five symbols.	**7**	(CO3, L3)

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16) a)	A university campus has several buildings that need to be interconnected by a network of pathways. Using graph theory, solve the following problem (i) - (iii): (i) Represent the campus layout as a graph where each building is a vertex and each pathway between buildings is an edge. Determine if this graph contains a spanning tree and construct it if possible. (ii) If a minimum cost spanning tree exists for the pathways, calculate its total cost given that the cost of each pathway is equal to its length in meters. (iii) Explain how Dijkstra's algorithm can be applied to find the shortest path from the main library (a specific building) to every other building on campus.	**7**	(CO3, L3)
	OR		
b)	A city planner needs to optimize a network of roads connecting five key areas: A, B, C, D, and E. The planner must ensure that every area can be reached from any other area using these roads while minimizing the total length of road construction. Describe how graph theory can be applied to this problem. Specifically, (i) represent the city layout as a weighted graph with vertices A, B, C, D, and E, and edges indicating possible roads between them; (ii) determine if the given network is Eulerian or Hamiltonian, justifying your answer; (iii) propose an optimal road network that ensures every area can be reached from any other area using the fewest number of roads. Use Konigsberg's bridge problem as a reference in your explanation.	**7**	(CO3, L3)
17)	Consider a university's online course scheduling problem where courses are represented as vertices and conflicts (e.g., overlapping time slots or shared students) between courses are represented as edges connecting those vertices, forming an undirected graph G with 10 courses and 25 conflicts. Determine whether the graph can be colored using four colors such that no two adjacent vertices share the same color. If it is possible, provide a coloring scheme. If not, explain why four colors are insufficient by discussing the chromatic number of the graph. (i) Construct the adjacency matrix for the given graph G. (ii) Use this matrix to determine if there exists a valid 4-coloring. (iii) Justify your answer using the properties of graphs and colorings.	**7**	(CO2, L3)
	Section D		**7*3=21**
18)	Construct a simple graph with 10 vertices such that each vertex has a degree of at least 4. Describe how you would determine if your constructed graph meets the given condition, and provide an adjacency list representation for your graph. (i) Explain what is meant by the degree of a vertex in a graph. (ii) Illustrate why your constructed graph satisfies the minimum degree requirement.	**7**	(CO2, L2)
19) a)	Analyze the use of graph theory in optimizing data storage for a library management system, considering (i) organizing books by subject areas and (ii) creating an efficient search algorithm to locate books.	**7**	(CO2, L4)
	OR		
b)	Analyze how to determine whether a given graph contains an Eulerian path or cycle. Differentiate between the conditions for a graph to have an Eulerian path and an Eulerian cycle. Provide examples of graphs that illustrate these concepts.	**7**	(CO2, L4)

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20)	Explain how graph theory can be applied to optimize data transmission in a network with 10 vertices. Consider a scenario where each vertex represents a node in the network, and the edges represent the communication links between them. Each link has an associated cost based on the quality of the connection (e.g., bandwidth, latency). Using this information, describe how you would use graph theory to determine the most efficient way to route data from a source node to a destination node. Additionally, calculate the average number of bits used per character if Huffman coding is applied to optimize the transmission of symbols with given frequencies: A: 0.20, B: 0.10, C: 0.15, D: 0.25, E: 0.30.	**7**	(CO1, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create